2020春高数下期末试题（回忆版）

1. Determine whether the following statements are true or false? No justification is necessary.

(1) Equation $r=2\sin(\theta)$ $(0\le\theta\le\pi)$ in polar form is a circle of radius 1 centered at $(0,1)$.

(2) If $f(x,y)=\sin x+\sin y$, then for any direction u, the directional derivative of $f(x,y)$ statisfies $-\sqrt{2}\le D_u f(x,y)\le\sqrt{2}$.

(3) If $u\neq 0$, and if $u\times v=u\times w$ and $u\cdot v=u\cdot w$, then $v=w$.

2. Multiple Choice Questions : (only one correct answer for each of the following questions.)

(1) Let $R=(x-1)^2+y^2\le 1$, then the integral $\iint_R f(x,y)\,dA$ is not equal to

(A) $\int_0^2\int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} f(x,y)\,dy\,dx$

(B) $\int_{-1}^{1}\int_{1-\sqrt{1-y^2}}^{1+\sqrt{1-y^2}} f(x,y)\,dx\,dy$.

(C) $\int_0^{2\pi}\int_0^1 f(1+r\cos\theta, r\sin\theta)\cdot r\,dr\,d\theta$

(D) $\int_0^{2\pi}\int_0^{2\cos\theta} f(r\cos\theta, r\sin\theta)\cdot r\,dr\,d\theta$.

(2) Which formula satisfies the conditions that function $f(x,y)$ has both partial derivaties at $(0,0)$ when $f(0,0)=0$?

(A) $\frac{xy}{x^2+y^2}$ (B) $\frac{x^2-y^2}{x^2+y^2}$ (C) $\sqrt{x^2+y^2}\sin\frac{1}{x^2+y^2}$ (D) $\frac{x^4+y^2}{x^2+y^2}$

(3) If $f(x,y)=3x+4y-ax^2-2ay^2-2bxy$ has only local maxima, then

(A) $2a^2>b^2$, and $a<0$.

(B) $2a^2>b^2$, and $a>0$.

(C) $2a^2<b^2$, and $a<0$.

(D) $2a^2<b^2$, and $a>0$.

3. Please fill in the blank for the questions below.

(1) If a plane is tangent to the surface $x^2-2y^2+z^2=2$, and parallel to $x-y+2z=0$, then the equation of the plane is _____.

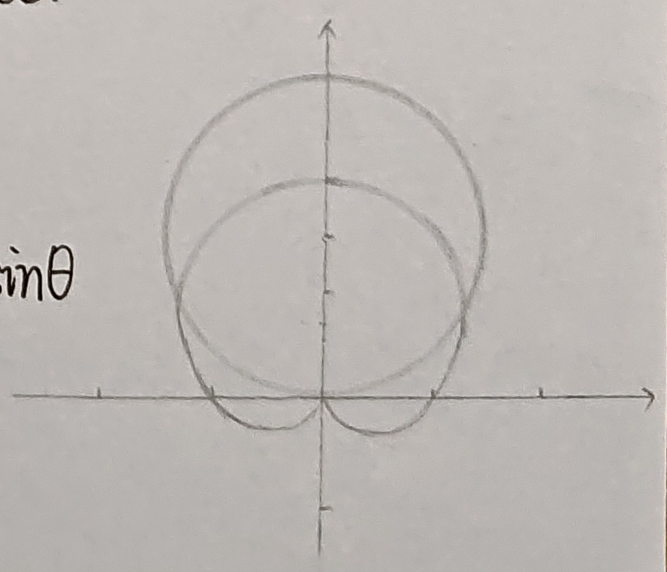
(2) Let $f(x,y,z)=(\frac{x}{y})^{\frac{1}{z}}$, then $df(1,1,1)=$ _____.

(3) The equation of the plane through the line $x=-1+2t$, $y=3+t$, $z=-t$ and parallel to the line $x=-2t$, $y=t$, $z=1-t$ is _____.

(4) The circulation of the field $F=\nabla(xy^2z^3)$ around the ellipse.

$$C: r(t)=(\cos t)\,i+(4\sin t)\,j,\ 0\le t\le 2\pi,$$

is _____.



4. Find the area of ~~region~~ region that lies inside the circle $r=3\sin\theta$ and outside the cardioid $r=1+\sin\theta$.

5. Find the points on the curve

$$r(t)=(12\sin t)\,i-(12\cos t)\,j+5t\,k.$$

at a distance 26π units along the curve from the point $(0,-12,0)$.

6. Find the interval of convergence of the power series $\sum_{n=0}^{\infty}\frac{2^n}{\ln(n+2)}x^n$.

7. Find the real numbers a,b $(b\neq 0)$, which satisfy

$$\lim_{x\to 0}\frac{\cos(\sin x)-\sqrt{1-x^2}}{x^a}=b.$$

8. Find the absolute maximum and minimum values of $f(x,y) = e^{-x^2-y^2}(x^2+2y^2)$ on the close disk $x^2+y^2 \leq 4$.

9. Evaluate the integal $\iiint z\sqrt{x^2+y^2+z^2}\, dV$, where D is the solid bounded above by $z=1$ and below by $z=\sqrt{x^2+y^2}$.

10. Calculate the line integral $\int_L \sin 2x\, dx + 2(x^2-1)y\, dy$, ~~bere~~ here L is the curve $y=\sin x$, from $(0,0)$ to $(\pi,0)$.

11. Use the Stokes' Theorem to calculate the circulation of the field F around the curve C in the indicated direction, here $F = yi + xzj + x^2k$, and C is the boundary of the triangle cut from the plane $x+y+z=1$ by the first octant, counterclockwise when viewed from above.

12. Use the Divergence Theorem to find the outward flux of F across the boundary of the region D, here $F = x^2i + y^2j + z^2k$; and D is the region cut from the solid cylinder $x^2+y^2 \leq 4$ by the planes $z=0$, and $z=1$.